

System Thinking and System Archetypes

ORSA Study text



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Sources:

MARTIN, Juan. *Vensim Online Course*. Online Course. 2024.

OpenAI. (2024). ChatGPT (Month Day version) [Large language model]. Available: <https://chat.openai.com>

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System Thinking and System Archetypes

IDENTIFYING THE PROBLEM

What is the problem?

We are going to learn a method for constructing simulation models that help us determine the best solution for a given problem. These are therefore management models, not predictive models.

Firstly, we have to identify the problem clearly and give a precise description of the aims of the study. It may be obvious, but it is very important that the definition of the problem be correct, since all further steps depend on this. This is also very useful when establishing the amount of time and money that will be spent creating the model.

Once the core of the problem is defined, a description must be completed, based on the knowledge of experts on the subject, basic documentation, etc. The result of this phase should be a preliminary perception of the elements that have a bearing on the problem, the hypothetical relationships between them, and their historical behaviour.

The historical reference of a system is a record of the historical behaviour of the main elements that are believed to influence the problem. Where possible, they should be quantified. This is the graphical and numerical representation of the verbal description of the problem.

The construction of a model is a long and costly process. It can't be justified if there are other more simple ways of obtaining the same results. There are essentially two other ways - statistics and intuition.

Statistics, or numerical calculus methods, are very useful for solving problems where there is an abundance of historical data or when we can assume reality will remain stable. For example, if you want to find out how many cars will drive past your house today, all you need is sufficient historical data and assuming the street hasn't changed, you'll get a good approximation.

Intuition has got you where you are today, so don't underestimate it. For many problems, intuition provides the right answers, drawing on our experience and knowledge. Intuition is cheap and fast. Keep using it as often as possible.

Only when we can't apply one of these two options with certainty must we resort to constructing a simulation model. Once the problem is defined, we will see that there are many directly or indirectly related aspects, or elements, which are also interrelated. They needn't be clearly or obviously interrelated. These elements constitute the system. We will now study reality as a system.

DEFINING THE SYSTEM

What is a system?

A system is a set of interrelated elements, where any change in any element affects the set as a whole. Only elements directly or indirectly related to the problem form the system under study here. In order to study a system, we must know the elements that make it up, and the relationships between them.

When we analyse a system we usually focus merely on the characteristics of its constituent elements. However, in order to understand the functioning of a complex system, we must focus also on the relationships that exist between the elements which form the system. It is impossible to understand the essence of a symphony orchestra by merely observing the musicians and their instruments. It is the coordination that exists between them that produces beautiful music.

A good method to begin defining a system is to write the main problem down in the middle of a blank page, surrounding it with the directly related elements. The elements which affect the main problem indirectly go around the appropriate direct elements. This will be the system that we will study in order to consider possible solutions to a given problem.

THE BOUNDARIES OF A SYSTEM

Where does the system end?

We have all heard the theory that a butterfly fluttering its wings in China could cause a tornado in the Caribbean. In our study, however, we will include only elements with a reasonable influence on the behaviour of a system. We mustn't lose sight of the objective, that is, to propose practical action towards effectively solving the problem at hand.

The system must contain as few elements as possible while providing a simulation that will truly allow us to decide which of the possible courses of action studied are the most effective solution to the problem. The models are generally small to begin with, with few elements. They are then expanded and perfected. Later on, elements which don't play a decisive part in the problem are eliminated. During the construction of a model, there are several extension and simplification phases in which elements are added and subtracted.

The final size of the model must be such that its main aspects can be explained in ten minutes. Any model larger than this will fail.

THE CAUSAL LOOP DIAGRAM

How do we represent the system?

The set of elements that bear relationship to the problem and that account for the observed behaviour, along with the relationships that exist between these constituent elements (which often involve feedback) form the system. The causal diagram represents the key elements of the system and the relationships between them.

As discussed above, it's important to draft versions that will bring us increasingly closer to the final complex model. The minimum set of elements and relationships that serves to reproduce the historical reference of a system is that which forms the basic structure of the system.

Once the variables of a system and the hypothetical relationships between these variables are known, we can move on to produce a graphical representation. This diagram shows the relationships as arrows between the variables. These arrows are marked with a sign (+ or -) which indicates the kind of influence one variable exerts over the other. A '+' means a change in the influencing variable will produce a change of the same direction in the target variable. A '-' means the effect will be the opposite.

So, when an increase in A results in an increase in B, or a fall in A causes a fall in B, this is a positive relationship, as shown below:

Positive does not mean good.



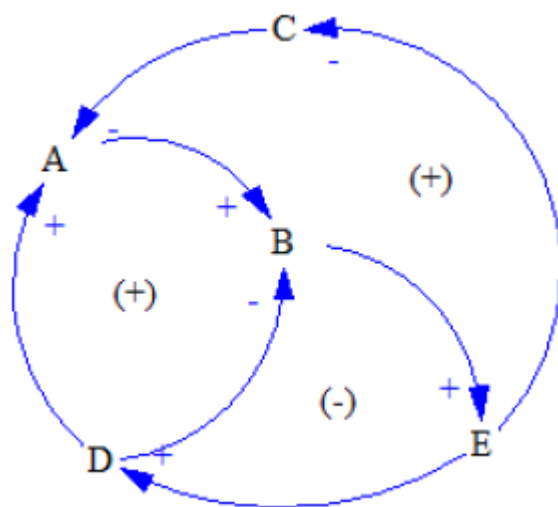
When an increase in A results in a fall in B, or a fall in A causes an increase in B, this is a negative relationship, which is expressed as follows:

Negative does not mean bad.



FEEDBACK LOOPS

What is a loop?



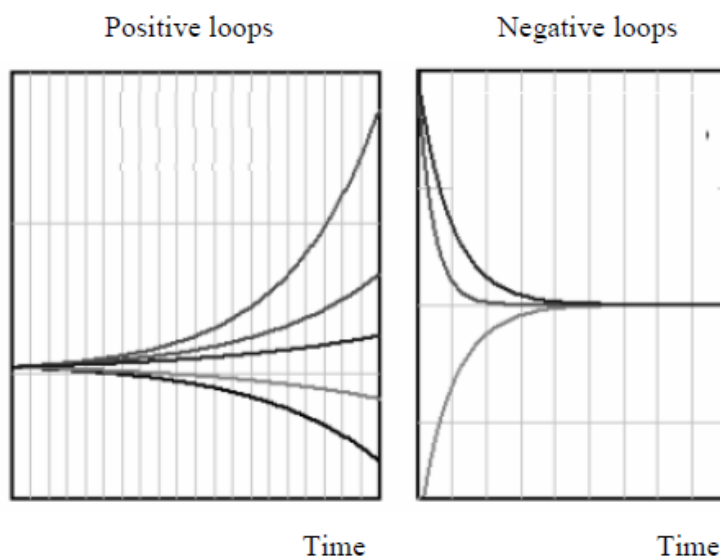
A closed chain of relationships is called a loop, or a feedback loop. When we turn on the tap to fill a glass with water, the amount of water in the glass increases. The amount of water in the glass, however, also has an effect on the speed at which it is filled. We fill it more slowly when it is fuller. Therefore, a loop exists.

The system formed by us, the tap and the glass is a negative loop, because it is designed to achieve a goal (fill the glass without spilling). Negative loops act as stabilising elements in systems designed to reach a given goal, like when a thermostat in a heating system guides the temperature towards the level specified by the user.

When we construct a model, loops appear. For example, those formed by ABEDA, DBED and ABECA in the following causal diagram.

Loops are defined as 'positive' when the number of negative relationships is even. If the number of negative relationships is odd, the loop is 'negative' (just as -3 multiplied by $+3$ gives -9).

Negative loops tend to stabilise the model, while positive loops tend to destabilise it, independently of the basic problem at hand.



Real-life systems contain both types of loops, and the ultimate behaviour will depend on the dominant type at any particular moment. When a country acquires more arms it makes its neighbours feel threatened and causes them to also acquire more arms. This is a positive loop, also called a vicious circle, and grows more and more as it feeds itself. Positive loops cause growth, and also the collapse of systems. Systems are made up of hundreds of interconnected positive and negative loops and its ultimate behaviour isn't obvious.

The concept of the feedback loop is very useful because it enables us to start from the structure of the system that we are analysing and work towards its dynamic behaviour. If a system fluctuates persistently, remains in equilibrium or drops off rapidly, we can identify the structural reasons and decide how to go about modifying the causal loops that are going to influence it.

Yet the most important use of this concept is in understanding **how the structure of systems affects their behaviour.**

CONTROL QUESTIONNAIRE PART 1

After reading this paragraph, it is advisable that the reader answer the following questionnaire to look at what is understood and what must be reviewed before continuing.

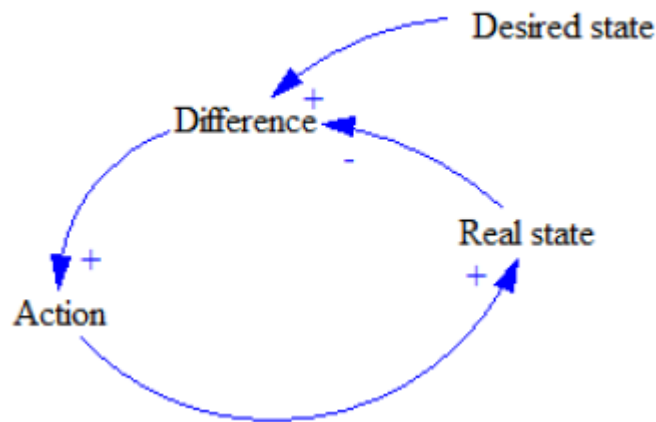
- a. Give some examples of **SYSTEM**. Remember the definition of system as a set of interrelated elements such that one element affects the behaviour of others. For example: a city.
- b. Name some **ELEMENTS** of the system. For example: persons, cars, pollution, streets, etc. Incorrect elements would be: the government, the city, Barcelona, colour, asphalt, etc. They are valid as elements of the system if we can notice when the element increases or decreases, improves or worsens, etc. Name the **UNITS** of the elements. For example: Persons: number of persons, Pollution: number of particles in suspension/m³, Streets: m².
- c. Give some examples of **CAUSAL RELATIONSHIPS** indicating whether they are positive or negative. For example: The more rainfall, the more people with umbrellas (positive).
- d. Give some examples of **FEEDBACK LOOPS**, which are formed where there is a 'closed circuit' between two or more elements of a system. For example: The warmer you are the fewer clothes you wear (negative relationship) and the more clothes you wear, the warmer you are (positive relationship). Both relationships together form a negative feedback loop.
- e. Exercise. Using PowerPoint, try defining a causal loop diagram from the following text. To do this, select the basic elements, link them and forget about numerical information.

Woodland is a prosperous region located in a temperate area with abundant vegetation. Its inhabitants lead a happy and comfortable life. Its total population is 1,000,000, a figure which has remained constant in recent years. Currently 40% of its population is young people under 20 years old, 50% are adults between 20 and 70 years old and the others are elderly people.

Furthermore, we know its population parameters to be the following - the life expectancy of the elderly people is 80 years, the birth rate of the adult population is 6% per year and the mortality rate is 2.5% in young people and 2% in adults.

We would like to make an estimation of how the total number of people in Woodland will evolve.

SYSTEM THINKING



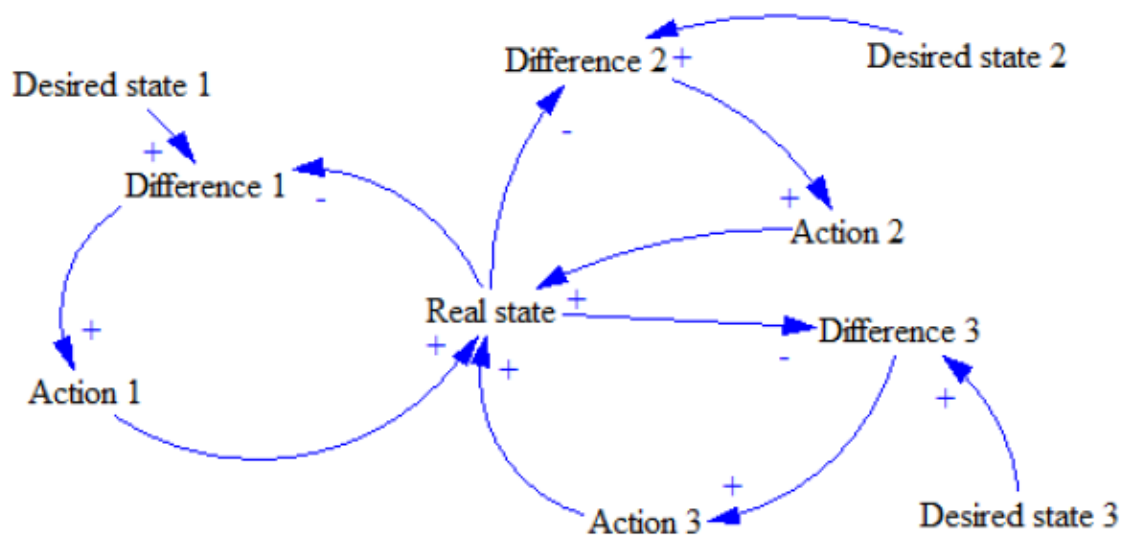
In complex systems, we can observe the same structure, that is, desired state - real state - difference - action, over and over again in very different contexts. On top of this base structure, generic structures have been identified that tend to appear regardless of the object of study.

There is always the same 'intelligent' structure that seeks to bring the real state closer to the desired state.

RESISTANCE TO CHANGE

When new managers join a firm, usually with new objectives, they often find that its employees put up resistance to everything they propose – 'They have already tried that. That won't work here. Our customers like it the way it's always been. That proposal is very risky.' In short, the company acts as a system that has managed to survive innumerable economic crises in the past, and as a structure, is capable of neutralising any change, whether from inside or outside, due to the multiple relationships between its members. Each pursues a different objective, yet as a whole they have succeeded in endowing the firm with stability, although that doesn't mean its position is necessarily the most efficient. For this reason it is often wise for new managers to seek the commitment of the general manager for their new objectives as a way of achieving a certain amount of strength and aligning the other elements in the company towards these objectives.

Many systems are not only resistant to new policies designed to improve their state (greater productivity, lower costs, etc.) but also show a persistent tendency to worsen it, despite the efforts to improve the situation.



EROSION OF OBJECTIVES

The action required to shift the real state towards the desired state always demands an effort and this effort in turn requires a consumption of time, energy, money, etc.

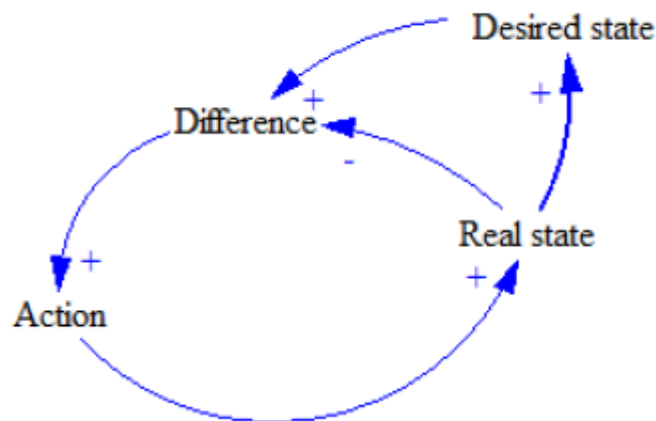
It is normal for the real state to 'contaminate' the desired state, that is, **for the system to try and avoid the consumption of energy required to take the action**. The desired state is initially reconsidered, since if it coincided with the real state, no action would be necessary. The diagram below shows this 'contamination' (the arrow from Real to Desired state).

If contamination occurs, the desired state is modified until it is the same as the real state. The difference is then zero and therefore there is no action to be taken. And so the real state of the system doesn't change.

The structure that brings about this behaviour is based on the idea that the system includes a particular objective (e.g., desired weight) that is compared with reality (real weight), and the discrepancy between these two values triggers an action which is proportional to the size of the gap. This is the usual pattern, seen up to now as a negative loop that tends to gradually pull the system towards its objective if it encounters some discrepancy. However, sometimes the state of the system can condition or modify the desired state, either because the real state is very long-lasting, because the action taken involves a great effort, or indeed for some other reason. The initial goal shifts towards the real state of the system.

This relieves the need to take action as the discrepancy has been reduced, not because the system has approached the objective but because the objective has approached the real state. As a result, the action taken is smaller.

In the case of the weight of obese people, this occurs when they accept that the target weight was too ambitious and that a more realistic target (a higher weight) is better. This argument serves as an excuse to follow a less strict diet. When they see their weight doesn't fall, they reconsider the ultimate target once again, and so on, until they think that actually their real weight is best, at which point they don't have to follow any sort of diet (this would have involved a sacrifice).



ADDICTION

Sometimes the real state of the system matches the desired state not as the result of action but due to support from outside the system. This support may or may not be permanent and may or may not be of interest, but the net effect is to bring the real state into line with the desired state, resulting in zero difference and therefore action by the system is unnecessary.

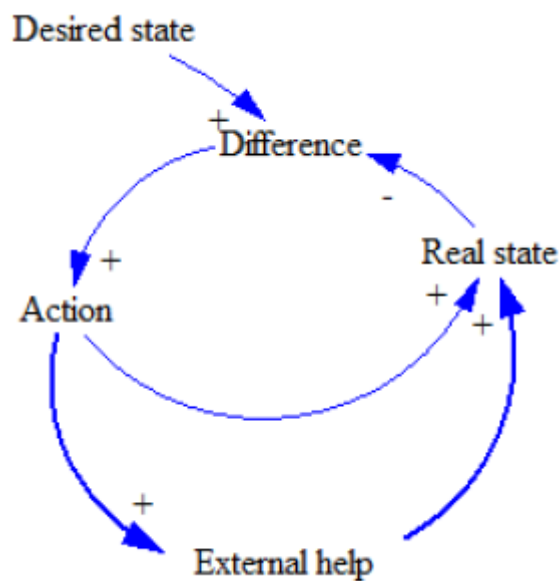
This phenomenon occurs when there is an objective that serves as a point of comparison with the state of the system. On the basis of the discrepancy observed, corrective measures are taken proportionally but in this case, the action taken doesn't serve to bring the system's real state closer to its desired one but rather to create the perception that the real system is close to the desired one, whereas in fact this action has no such effect.

The lack of clear perception of the real system leads to a situation in which the necessary corrective measures aren't taken because the state of the system is perceived as being closer to the objective than it really is.

When the immediate or short-term effect of the action disappears, the problem (i.e., the discrepancy between the real state and the desired one) reappears, often with greater intensity, so the system reapplies some measure that appears to solve the problem whenever the effect of the previous measure starts to fade.

Alcohol, nicotine and caffeine are obvious examples of addictive substances. Another case that springs to mind is the use of pesticides, which eliminate, together with the pest in question, the natural control mechanisms. As a result, the pest will reappear as soon as the effect of the pesticide abates, but this time without any natural control.

In cases of addictive systems, it is difficult to find suitable policies, since the action taken offers apparent results in the short term, but once the process is rolling, it is difficult to stop. Obviously, the best approach is to be aware of these types of processes, in other words, to be wary of using measures that attack the symptoms but make the system worse when they are relaxed. Once the addictive process has been started, you have to expect at least short-term difficulties if you plan to stop this process, be it physical pain for somebody who takes an addictive drug, rising petrol prices on inclusion of the associated environmental costs, or more pests and lower-quality food until such time as natural predators return.



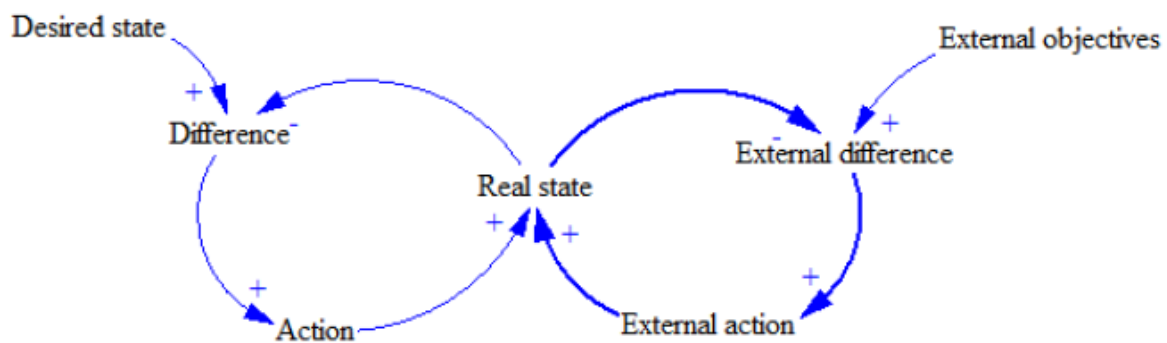
SHIFTING THE BURDEN TO THE EXTERNAL FACTOR

As they get older and spend more and more time reading, some people gradually get poorer eyesight. In the end they can't read what is written on a blackboard, and can't renew their driving licence, so they get glasses or contact lenses. Then, in one year their eyesight worsens as much as it did in the previous 30 years. So their glasses become a necessity not only to see at a distance but also to read a document. Apparently this happens because for years, the muscles around the eyes have been straining to compensate their poor vision and when this effort is no longer necessary, they cease to act and end up losing this ability totally. Before long, they need stronger lenses.

This is a classic example of shifting the burden to the external factor. In this sort of system an external force keeps the system in the desired situation. A well-intentioned, benevolent and very effective force decides to help us to get the system where we want it. This new mechanism works very well.

But with this process, through the active destruction of the impediments that redirected the system towards the desired position, or simply through atrophy, the original forces that worked to correct the position of the system are weakened. When the system moves away from the desired position the external factor makes an extra effort, which weakens the original forces still further. In the end the original system adopts a position of total dependence on the external factor, as its original corrective forces have disappeared completely and in most cases, irreversibly.

Seeking the aid of an external factor to get the system where we want it to be isn't in itself a bad thing. Usually, it is beneficial and enables the system to better tackle objectives. The dynamics of the system can be problematical for two reasons. Firstly, the external factor that intervenes doesn't tend to perceive the consequences of its help on the elements of the system, particularly on those that performed the same task as itself. Secondly, the system doesn't think that this help is temporary; they lose their long-term perspective and so become more vulnerable and dependent on the external factor.

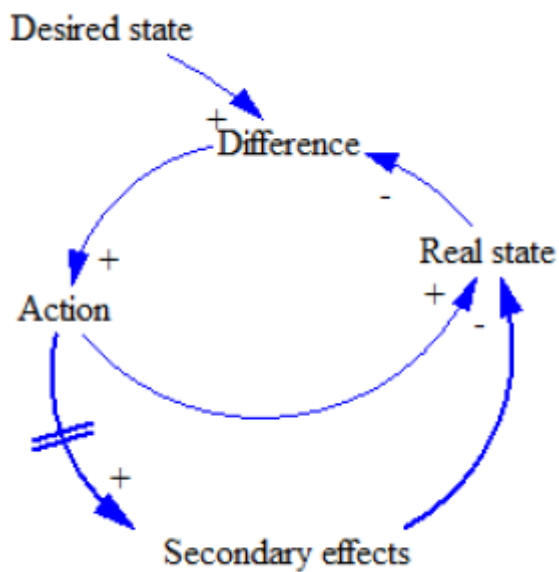


SHORT AND LONG-TERM EFFECTS

A rational analysis of the problem at hand based on our capacity for synthesis and our ability to imagine things seems to be a bad guide to finding the key factors. We generally pay attention to the components of the system and their behaviour in the short term, all on the basis of incomplete information. Consequently, firms reduce their stocks of finished products when sales are seen to slump, the government extends its tax reductions for small farmers, and policies are introduced to encourage firms to replace their machinery instead of maintaining it properly. They are all very reasonable policies. But there's still something inside us that just might make us realise that our customers' dissatisfaction with our long delivery schedules, or farmers' permanent concern with increasing their output, or the idea of replacing a machine that's productive, all means something, but we haven't given it the right interpretation.

Finally, I'd like to say that in my opinion, we have the capacity to understand not only simple systems but complex ones too, and to find the key factors. What we don't appear to have is the capacity to articulate the arguments to convince others or even ourselves that what we're perceiving is right. We expect the solution to be closely related to the symptom; we expect long-term profit to start with short-term profit, or a strategy that's satisfactory for all the agents involved. Yet we know complex systems don't behave that way. So something inside us still insists somehow that maybe that simple, effective solution isn't the best. And then we carry on proposing policies that can't work, denying ourselves other simpler and more effective ones that could.

We try to compete instead of cooperating, and we try to reach the limits of the environment's capacity instead of admitting that we have already gone too far. The results are famine, war, pollution and depression. And right in front of us, within reach of our capacity for understanding, stand balance between countries, peace, equality and sustainable development.



CONTROL QUESTIONNAIRE PART 2

After reading this paragraph, it is advisable that the reader answer the following questionnaire to look at what is understood and what must be reviewed before continuing.

- Give a system example that has **JUST ONE GOAL**, indicating the goal. For example: a mower; goal: to cut grass.
Give a system example that has **SEVERAL GOALS**. For example: a company, where the businessperson has the following objectives: the most profit, increasing the number of clients and increasing product quality.
- Give a system example that has **GOAL EROSION**. Indicate some ways to avoid contamination (in other words, erosion) of the goal by the real situation by securing it to an external element. For example: students usually have an erosion of their initial goal of getting excellent grades. In this case, an external element that can prevent this erosion is the grades of a 'rival' student.
- Give a system example that shows **RESISTANCE TO CHANGE**. For example: We prefer to wear our old shoes because they are more comfortable than new ones.